
$\Sigma 1$: ONTIC MANIFOLD SCALING AND APPROXIMATE PROJECTABILITY IN THE LARGE- N REGIME

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ABSTRACT

Geometric formulations of finite-dimensional quantum theory describe dynamics on projective state space $\mathbb{CP}^{N-1}$, but do not by themselves specify when deeper ontic dynamics should reduce to an effectively closed flow on that space. In the finite-dimensional reconstruction developed in earlier work, this role is played by the approximate projectability condition, which identifies the regime in which ontic Hamiltonian dynamics induces controlled projected Hamiltonian flow on epistemic space, with Schrödinger-form evolution recovered when the effective Hamiltonian lies in the operator-induced quadratic class Blore [2026g].

In the present paper, we give the first systematic large- N treatment of this condition. We introduce a family of finite-dimensional compact Kähler ontic manifolds

$$(\Sigma_N, \omega_{\Sigma}^{(N)}, J_{\Sigma}^{(N)}, g_{\Sigma}^{(N)}, \mu_L^{(N)}, \Psi_t^{(N)})$$

together with compatible projections $\pi_N : \Sigma_N \rightarrow \mathbb{CP}^{N-1}$, and define an admissible class of Hamiltonians for which approximate projectability holds in a controlled sense. The analysis is explicitly regime-based: approximate projectability is formulated as a scale-dependent property, characterised by bounds on fibre variation and by stability under coarse-graining and timescale separation Blore [2026g,m,d].

Conceptually, approximate projectability is identified with the projectability form of physical isolation: a subsystem admits a quantum-effective description when fibre-sensitive components of the ontic Hamiltonian remain below operational resolution over the relevant timescale. This replaces a bare selection of projectable dynamics with a controlled effective-isolation criterion, without claiming a universal first-principles derivation of the projection structure.

Rather than claiming a universal derivation, the paper formulates a disciplined bridge framework. It identifies sufficient conditions under which projected dynamics remains well defined and approximately closed on $\mathbb{CP}^{N-1}$ over operational timescales, and clarifies how this admissible regime supports later continuum-like and locality-facing work. No claim is made that all physically relevant Hamiltonians satisfy these conditions, nor that a full continuum, relativistic, or field-theoretic reconstruction has been achieved. The paper therefore serves as a bridge within the present programme: it clarifies the status of approximate projectability, replaces a bare selection principle with a controlled effective framework, and specifies the role of this condition in connecting exact finite-dimensional models to large- N analysis Blore [2026d].

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1 Introduction

1.1 Purpose of the Present Paper

Geometric formulations of finite-dimensional quantum theory describe dynamics as Hamiltonian flow on projective state space $\mathbb{CP}^{N-1}$, equipped with its canonical symplectic and Kähler structure (Kibble 1979; Ashtekar & Schilling 1999). This formulation provides a complete and internally consistent account of quantum kinematics and unitary evolution. However, it does not by itself specify when or why a deeper ontic dynamics, defined on a more detailed state space, should reduce to an effectively closed flow on $\mathbb{CP}^{N-1}$.

In finite-dimensional deterministic reconstructions, this issue appears as a structural condition on the underlying Hamiltonian dynamics. In particular, for projected dynamics to be well defined, variation of the ontic Hamiltonian along the fibres of the projection map must be sufficiently small. This requirement is referred to as *approximate projectability*. At fixed finite dimension, it identifies a quantum-effective regime in which projected dynamics is effectively closed on projective state space. Schrödinger-form evolution is recovered when the effective Hamiltonian lies in the operator-induced quadratic class Blore [2026g].

The purpose of the present paper is to give the first systematic treatment of approximate projectability beyond fixed finite dimension. Rather than treating it as a selection rule imposed on admissible Hamiltonians, we formulate it as a controlled, scale-dependent property within a family of finite-dimensional models. The goal is to define the regime in which approximate projectability holds, to characterise the associated admissible class of ontic dynamics, and to clarify how this regime supports subsequent large- N and continuum-like developments.

In geometric quantum mechanics, Hamiltonian flow on $\mathbb{CP}^{N-1}$ is taken as fundamental. The present work instead treats such flow as an effective description arising under controlled projection from deeper ontic dynamics.

The central question addressed in this paper is when deterministic ontic Hamiltonian flow on Σ_N induces an approximately closed dynamics on the projected state space $\mathbb{CP}^{N-1}$. This is the projectability problem.

The physical interpretation is effective isolation. A subsystem admits a quantum-effective description only when the hidden variation within the fibres of

$$\pi_N : \Sigma_N \rightarrow \mathbb{CP}^{N-1}$$

does not produce operationally resolvable deviations in the projected dynamics. Exact projectability corresponds to ideal isolation: all ontic representatives of the same projected ray induce the same motion on $\mathbb{CP}^{N-1}$. Approximate projectability corresponds to imperfect but operationally stable isolation: fibre-sensitive variation is present, but its accumulated projected effect remains below the resolution at which the quantum-effective description is being used.

Thus the purpose of the present paper is not to claim that every ontic Hamiltonian is projectable. It is to define the controlled regime in which projected Hamiltonian dynamics is valid, and to state explicitly when that regime fails.

1.2 The Approximate Projectability Problem

Consider a family of finite-dimensional ontic manifolds Σ_N equipped with Liouville-preserving Hamiltonian flows $\Psi_t^{(N)}$, together with smooth projection maps

$$\pi_N : \Sigma_N \rightarrow \mathbb{CP}^{N-1}.$$

The projection π_N identifies equivalence classes of ontic microstates that correspond to the same epistemic state. The fibres of π_N therefore represent directions in which ontic variation is operationally invisible at the level of projective state space.

For a general Hamiltonian H_N on Σ_N , the induced flow need not descend to a well-defined vector field on $\mathbb{CP}^{N-1}$. The obstruction arises from variation of H_N along the fibres of π_N . If this variation is uncontrolled, two ontic trajectories projecting to the same epistemic state may evolve into distinct projected states, preventing the existence of a closed effective dynamics on $\mathbb{CP}^{N-1}$.

Approximate projectability provides a controlled way to avoid this obstruction. At finite N , it is formulated as a small-variation condition along fibres, ensuring that projected dynamics remains well defined up to bounded error over operational timescales Blore [2026g]. The resulting projected flow is then approximately Hamiltonian with respect to the Fubini–Study structure. The standard Schrödinger form is recovered when the effective Hamiltonian belongs to the operator-induced quadratic class.

The central problem addressed in this paper is how to extend this condition beyond fixed finite dimension. Specifically:

- Under what structural and dynamical conditions does approximate projectability persist across a family (Σ_N, π_N) ?
- How should the associated error bounds scale with N ?
- What class of Hamiltonians should be regarded as admissible for participation in large- N or continuum-like limits?

1.3 Relation to Finite-Dimensional Reconstruction

In earlier work, finite-dimensional models provide a sector-conditional reconstruction of the operational structure of finite-dimensional quantum mechanics. Ontic dynamics is given by Hamiltonian flow on a compact manifold Σ , while observable statistics arise after projection to $\mathbb{CP}^{N-1}$ together with an invariant measure structure and an operational probability rule derived from symmetry considerations Blore [2026a,g].

Within that setting, approximate projectability appears as a condition defining the quantum-effective sector. When it holds, the projected dynamics on $\mathbb{CP}^{N-1}$ is approximately closed and reproduces the standard unitary evolution. When

it fails, the projected dynamics is no longer well defined, and the system lies outside the regime described by standard quantum mechanics.

The present paper does not modify this finite-dimensional reconstruction. Instead, it extracts and reformulates the approximate projectability condition in a form suitable for comparison across different truncation levels. In this sense, it provides a structural extension rather than a new dynamical postulate.

1.4 Relation to Controlled Continuum-Like Limits

A separate line of development studies sequences of finite-dimensional models and their convergence properties as $N \rightarrow \infty$. In that framework, all statements remain exact at each finite N , and continuum-like behaviour is introduced only through controlled asymptotic comparison of finite- N statistics.

The present work is complementary to that analysis. Whereas convergence results address the behaviour of outcome statistics and observables, the focus here is on the dynamical structure required for those comparisons to be meaningful. In particular, the existence of a well-defined projected dynamics at each N is a prerequisite for interpreting large- N limits in terms of effective quantum evolution.

Accordingly, this paper should be understood as providing the admissibility conditions required for the convergence framework. It specifies which classes of ontic Hamiltonians are compatible with the existence of a quantum-effective regime across scales, without assuming that all physically relevant systems satisfy these conditions.

1.5 Main Claims and Non-Claims

The results of this paper are deliberately limited in scope.

Claims.

- A definition of approximate projectability as a controlled, scale-dependent condition on ontic Hamiltonians.
- The identification of an admissible class of Hamiltonians for which projected dynamics remains well defined up to bounded error over operational timescales.
- A formulation of approximate projectability as a regime property rather than a universal structural constraint.
- A framework for analysing how this regime behaves across families (Σ_N, π_N) .

Non-claims.

- No claim is made that all physically realised Hamiltonians satisfy approximate projectability.
- No universal derivation of the quantum-effective sector is provided.
- No infinite-dimensional ontic manifold is constructed.
- No relativistic, local, or field-theoretic reconstruction is attempted.

The paper is therefore a bridge construction. Its role is to clarify the status of approximate projectability, replace a selection principle with a controlled effective framework, and define the admissible regime required for subsequent large- N and continuum-like developments.

1.6 Structure of the Paper

Section 2 reviews the finite-dimensional background and formulates the approximate projectability condition in precise terms. Section 3 develops the regime-based interpretation of this condition, including error bounds and stability properties. Section 4 defines admissible classes of Hamiltonians and analyses their structural features. Section 5 studies the resulting projected dynamics and its relation to Schrödinger-form evolution. Section 6 examines scaling behaviour across families (Σ_N, π_N) and its compatibility with convergence requirements. Section 7 discusses the role of these results in enabling continuum-like and locality-facing extensions. Section 8 summarises the scope and limitations of the framework, and Section 9 concludes.

2 Finite-Dimensional Background and the Projectability Condition

2.1 Ontic Manifolds and Hamiltonian Flow

For each finite truncation level N , the ontic state space is taken to be a compact, connected Kähler manifold

$$(\Sigma_N, \omega_\Sigma^{(N)}, J_\Sigma^{(N)}, g_\Sigma^{(N)}).$$

Here $\omega_\Sigma^{(N)}$ is a closed non-degenerate symplectic form, $J_\Sigma^{(N)}$ is an integrable complex structure, and $g_\Sigma^{(N)}$ is the associated Kähler metric satisfying

$$g_\Sigma^{(N)}(u, v) = \omega_\Sigma^{(N)}(u, J_\Sigma^{(N)}v)$$

for tangent vectors $u, v \in T_x \Sigma_N$.

The Liouville measure on Σ_N is defined by the symplectic volume form

$$\mu_L^{(N)} = \frac{(\omega_\Sigma^{(N)})^{d_N}}{d_N!}, \quad d_N := \frac{1}{2} \dim_{\mathbb{R}} \Sigma_N.$$

Compactness ensures that

$$\mu_L^{(N)}(\Sigma_N) < \infty.$$

A smooth Hamiltonian

$$H_N : \Sigma_N \rightarrow \mathbb{R}$$

defines a Hamiltonian vector field X_{H_N} by

$$\iota_{X_{H_N}} \omega_\Sigma^{(N)} = dH_N.$$

The corresponding ontic flow is denoted

$$\Psi_t^{(N)} : \Sigma_N \rightarrow \Sigma_N.$$

By Liouville's theorem, this Hamiltonian flow preserves the Liouville measure:

$$(\Psi_t^{(N)})_* \mu_L^{(N)} = \mu_L^{(N)}.$$

Thus the primitive finite- N ontic data are

$$(\Sigma_N, \omega_\Sigma^{(N)}, J_\Sigma^{(N)}, g_\Sigma^{(N)}, \mu_L^{(N)}, H_N, \Psi_t^{(N)}).$$

The projection to epistemic state space is additional structure, introduced below. It is not part of the definition of Σ_N itself.

2.2 Projection to Epistemic State Space

The operational description is obtained via a smooth surjective map

$$\pi_N : \Sigma_N \rightarrow \mathbb{CP}^{N-1},$$

which assigns to each ontic microstate an epistemic state. The space $\mathbb{CP}^{N-1}$ carries its canonical Kähler structure, including the Fubini–Study metric and symplectic form.

The projection π_N induces a partition of Σ_N into fibres

$$F_{[\psi]}^{(N)} := \pi_N^{-1}([\psi]), \quad [\psi] \in \mathbb{CP}^{N-1},$$

which represent equivalence classes of ontic states that are operationally indistinguishable at the epistemic level.

Pushforward of the Liouville measure defines an epistemic measure on $\mathbb{CP}^{N-1}$, which in the finite-dimensional setting is taken to be absolutely continuous with respect to the Fubini–Study measure. This ensures compatibility between ontic volume structure and operational probability assignments Blore [2026a].

2.3 Projected Dynamics and the Closure Problem

Given an ontic trajectory $x(t) = \Psi_t^{(N)}(x_0)$, one may define the projected trajectory

$$\psi(t) = \pi_N(x(t)).$$

In general, however, the map $x_0 \mapsto \psi(t)$ does not define a closed dynamical system on $\mathbb{CP}^{N-1}$. The obstruction arises because different ontic representatives of the same epistemic state may evolve differently under the Hamiltonian flow. More precisely, if $x, x' \in \Sigma_N$ satisfy $\pi_N(x) = \pi_N(x')$, then the corresponding trajectories $x(t)$ and $x'(t)$ need not project to the same $\psi(t)$. In that case, the projected evolution depends on the choice of representative within the fibre, and no well-defined vector field exists on $\mathbb{CP}^{N-1}$.

This is the *closure problem*: determining when the ontic dynamics induces a consistent, representative-independent evolution on epistemic state space.

2.4 Exact and Approximate Projectability

A Hamiltonian H_N is said to be *exactly projectable* if its Hamiltonian vector field X_{H_N} is π_N -related to a vector field X_h on $\mathbb{CP}^{N-1}$. In that case, the diagram

$$\begin{array}{ccc} \Sigma_N & \xrightarrow{\Psi_t^{(N)}} & \Sigma_N \\ \downarrow \pi_N & & \downarrow \pi_N \\ \mathbb{CP}^{N-1} & \xrightarrow{\phi_t} & \mathbb{CP}^{N-1} \end{array}$$

commutes, and the projected dynamics is exactly closed.

Exact projectability is, however, a strong condition and is not expected to hold for generic Hamiltonians. Instead, one considers *approximate projectability*, in which the failure of exact closure is controlled.

Formally, approximate projectability is expressed as a small-variation condition along fibres. Let $V_x = \ker(d\pi_N)_x$ denote the vertical subspace at $x \in \Sigma_N$. The Hamiltonian H_N is approximately projectable on a region $S \subset \Sigma_N$ if

$$\sup_{x \in S} \|dH_N|_{V_x}\| \leq \varepsilon_N,$$

for some small parameter ε_N .

Under this condition, the variation of the Hamiltonian along fibres is uniformly bounded, and the projected dynamics is well defined up to controlled error over finite timescales.

2.5 Approximate Closure of Projected Dynamics

When approximate projectability holds, one may define an effective projected trajectory $y(t)$ on $\mathbb{CP}^{N-1}$ generated by a Hamiltonian h_N satisfying

$$h_N \circ \pi_N \approx H_N.$$

The true projected trajectory $\psi(t) = \pi_N(x(t))$ then remains close to $y(t)$ in the Fubini–Study metric. Specifically, for finite operational times $t \in [0, T]$, one expects a bound of the form

$$d_{FS}^{(N)}(\psi(t), y(t)) \leq C_T^{(N)} \varepsilon_N,$$

where $C_T^{(N)}$ depends on the time horizon, the projection geometry, and regularity properties of the flow.

This establishes an *approximate closure* of the projected dynamics: although the evolution is not exactly representative-independent, the deviation remains controlled and operationally negligible within the regime defined by ε_N .

2.6 Interpretation and Role in Finite Models

At fixed finite N , approximate projectability identifies a regime in which the projected dynamics on $\mathbb{CP}^{N-1}$ behaves as an effectively closed Hamiltonian system. In this regime, the induced flow is approximately compatible with the Fubini–Study structure. Schrödinger-form evolution is recovered when the effective Hamiltonian belongs to the operator-induced quadratic class Blore [2026g].

Outside this regime, the projected dynamics is not well defined, and no consistent quantum-effective description exists. The system must then be treated at the ontic level without reduction to projective dynamics.

The approximate projectability condition therefore plays a structural role: it delineates the domain in which finite-dimensional ontic models admit an effective quantum description. The aim of the present paper is to reformulate this condition in a way that remains meaningful across families (Σ_N, π_N) and to determine how it behaves under scaling to large N .

2.7 Core Definitions

Definition 2.1 (Vertical subspace). $V_x := \ker(d\pi_N)_x$.

Definition 2.2 (Approximate projectability). A Hamiltonian H_N is approximately projectable on $S \subset \Sigma_N$ if

$$\sup_{x \in S} \|dH_N|_{V_x}\| \leq \varepsilon_N.$$

Definition 2.3 (Admissible Hamiltonian). H_N is admissible on (S, T, η_N) if it satisfies approximate projectability on S , yields controlled projected Hamiltonian dynamics over $t \in [0, T]$, and satisfies

$$C_T^{(N)} \epsilon_N \ll \eta_N.$$

Definition 2.4 (Schrödinger-admissible Hamiltonian). An admissible Hamiltonian H_N on (S, T) is Schrödinger-admissible if its effective projected Hamiltonian h_N lies in the operator-induced quadratic class. Equivalently, there exists a Hermitian operator $\hat{H}_N$ on the finite-dimensional Hilbert representative such that

$$h_N([\psi]) = \frac{\langle \psi | \hat{H}_N | \psi \rangle}{\langle \psi | \psi \rangle}.$$

In this case, the projected Hamiltonian flow on CP^{N-1} is the projectivisation of finite-dimensional Schrödinger evolution.

3 Approximate Projectability as a Regime Condition

3.1 Fibre Structure and Hidden Variation

The projection map

$$\pi_N : \Sigma_N \rightarrow CP^{N-1}$$

canonically defines a vertical subspace at each point $x \in \Sigma_N$. A horizontal complement may be chosen locally when needed, but only the vertical subspace is fixed by the projection itself. The vertical subspace

$$V_x := \ker(d\pi_N)_x$$

consists of directions along which ontic variation is invisible at the epistemic level. These directions parameterise the internal structure of the fibre

$$F_{[\psi]}^{(N)} = \pi_N^{-1}([\psi]).$$

Variation of the Hamiltonian along these vertical directions determines whether the ontic dynamics can be consistently projected. If the Hamiltonian varies significantly along fibres, then ontic trajectories that are initially epistemically indistinguishable may diverge under projection. This leads to a breakdown of closure on $\mathbb{CP}^{N-1}$.

Approximate projectability therefore requires that the Hamiltonian be approximately constant along fibres, in a sense made precise below.

3.2 Definition of Approximate Projectability

Let $S \subset \Sigma_N$ be a fixed open set with compact closure $\bar{S}$. We assume that S contains the support of the relevant preparation measure and that trajectories generated by the Hamiltonian flow remain in S for all $t \in [0, T]$. Thus S is chosen prior to the admissibility analysis as a controlled geometric region on which the required smoothness and boundedness estimates hold. The Hamiltonian H_N is said to be approximately projectable on S if there exists a parameter $\varepsilon_N > 0$ such that

$$\sup_{x \in S} \|dH_N|_{V_x}\| \leq \varepsilon_N.$$

Equivalently, for any pair of points $x, x' \in S$ satisfying $\pi_N(x) = \pi_N(x')$, one has

$$|H_N(x) - H_N(x')| \leq C \varepsilon_N,$$

for some constant C depending on the geometry of the fibres.

This condition ensures that variation of the Hamiltonian along fibres is uniformly small on the relevant region. In the limit $\varepsilon_N \rightarrow 0$, one recovers exact projectability.

3.3 Error Bounds and Control Parameters

The parameter ε_N serves as a quantitative measure of the failure of exact projectability. Its operational significance is determined by how it propagates under the Hamiltonian flow.

Let $x(t) = \Psi_t^{(N)}(x_0)$ be an ontic trajectory with initial condition $x_0 \in S$, and let $\psi(t) = \pi_N(x(t))$ be the corresponding projected trajectory. Under approximate projectability, one may define an effective projected trajectory $y(t)$ generated by a Hamiltonian h_N on $\mathbb{CP}^{N-1}$.

For finite times $t \in [0, T]$, the deviation between the true and effective projected trajectories is controlled by an inequality of the form

$$d_{FS}^{(N)}(\psi(t), y(t)) \leq C_T^{(N)} \varepsilon_N,$$

where $d_{FS}^{(N)}$ denotes the Fubini–Study distance and $C_T^{(N)}$ depends on the regularity of the flow, the projection geometry, and the time horizon.

The pair $(\varepsilon_N, C_T^{(N)})$ therefore defines the regime of validity of approximate projectability. In particular, for fixed operational timescale T , the condition $C_T^{(N)} \varepsilon_N \ll 1$ ensures that deviations from exact closure remain negligible.

To give the smallness condition operational content, introduce a resolution scale

$$\eta_N > 0.$$

For a measurement context M , coarse-graining class $\mathcal{C}$, and statistical distinguishability threshold $\delta_{\text{op}} > 0$, η_N denotes the smallest Fubini–Study displacement on $\mathbb{CP}^{N-1}$ that produces an operationally distinguishable change in the relevant outcome statistics.

One possible formalisation is

$$\eta_N(M, \mathcal{C}, \delta_{\text{op}}) = \inf \left\{ d_{FS}^{(N)}([\psi], [\phi]) : \|p_M(\cdot | [\psi]) - p_M(\cdot | [\phi])\|_{\text{TV}} \geq \delta_{\text{op}} \right\},$$

where $p_M(\cdot | [\psi])$ is the probability distribution over outcomes for context M , and $\|\cdot\|_{\text{TV}}$ is total variation distance.

The quantum-effective regime is therefore not characterised only by formal smallness of the projectability error. It is characterised by the operational condition

$$C_T^{(N)} \varepsilon_N \ll \eta_N.$$

This states that the accumulated projected deviation generated by fibre-sensitive ontic dynamics is below the resolution of the epistemic description.

3.4 Stability Under Coarse-Graining

In practice, the relevant description of a system is often coarse-grained. Observables are insensitive to fine-scale structure within fibres, and preparation procedures define regions rather than individual microstates.

A key requirement for approximate projectability to be physically meaningful is stability under such coarse-graining. Let ρ_N be an epistemic density obtained as the pushforward of a Liouville-supported distribution on Σ_N . If approximate projectability holds, then the evolution of ρ_N under projected dynamics should remain close to that generated by the effective Hamiltonian h_N .

This stability ensures that the regime defined by ε_N is not an artefact of fine-grained structure, but persists under the operational descriptions relevant to experiments.

3.5 Timescale Separation and Effective Closure

Approximate projectability is inherently a dynamical notion. Even if variation along fibres is nonzero, its effects may remain small over sufficiently short timescales.

Two mechanisms contribute to effective closure:

- **Timescale separation:** If fibre-sensitive dynamics evolves on timescales much shorter or much longer than those relevant to observation, its net effect on projected dynamics may be suppressed.
- **Averaging:** Rapid oscillations along fibres may average out, leading to an effective Hamiltonian that depends only on the projected state.

These mechanisms do not eliminate fibre-sensitive dynamics at the ontic level. Rather, they render it operationally negligible within a specified time window. Approximate projectability should therefore be understood as a property relative to a chosen operational timescale T .

The mechanisms invoked here are structurally analogous to standard adiabatic and averaging arguments in both quantum and classical mechanics. In Born–Oppenheimer theory, rapidly adjusting degrees of freedom are averaged out to yield an effective slow-sector dynamics; in Kato’s adiabatic framework, slow parameter variation preserves an effective subspace structure; and in Hamiltonian averaging theory, rapidly oscillating variables contribute only controlled corrections to the slow flow (Born and Oppenheimer 1927; Kato 1950; Arnold 1989).

The present setting differs in two respects. First, the separation is formulated relative to the projection

$$\pi_N : \Sigma_N \rightarrow CP^{N-1},$$

so that the relevant distinction is between fibre-sensitive and projected dynamics rather than between canonical slow and fast coordinates on a single phase space. Second, the aim is not merely reduction at fixed dimension, but a scaling admissibility criterion across families of finite-dimensional models. The present framework therefore uses adiabatic-style ideas in a projective-geometric and bridge-oriented setting rather than reproducing classical averaging theory unchanged.

3.6 Regime Interpretation

The preceding analysis leads to a reinterpretation of approximate projectability. It is not a universal structural constraint on all Hamiltonians, but a *regime condition* defined by the smallness of fibre variation relative to operational scales.

More precisely, for a region $S \subseteq \Sigma_N$, timescale T , and operational resolution η_N , the quantum-effective regime consists of pairs (H_N, S) for which there exists a projectability-error parameter ϵ_N satisfying

$$C_T^{(N)} \epsilon_N \ll \eta_N.$$

Within this regime, projected dynamics is compatible with closed Hamiltonian flow on projective state space to the relevant operational accuracy. Schrödinger-form evolution is recovered when the effective Hamiltonian h_N lies in the operator-induced quadratic class.

The condition does not require fibre-sensitive ontic dynamics to vanish. It requires only that its accumulated projected effect be smaller than the resolution at which the quantum-effective description is being used.

Physical concept	Projectability expression
Effective physical isolation	Small fibre-sensitive residual ϵ_N
Autonomous subsystem dynamics	Approximately closed projected flow on $\mathbb{CP}^{N-1}$
Loss of effective isolation	Growth of $C_T^{(N)} \epsilon_N$
Operationally stable quantum behaviour	$C_T^{(N)} \epsilon_N \ll \eta_N$
Breakdown of projected closure	$C_T^{(N)} \epsilon_N \geq \eta_N$

Table 1: Approximate projectability as the mathematical form of effective physical isolation.

3.7 Summary

Approximate projectability is therefore best understood as a controlled, scale-dependent property of ontic Hamiltonians. It is characterised by:

- bounded variation of the Hamiltonian along fibres,
- controlled propagation of errors under dynamical evolution,
- stability under coarse-graining, and
- dependence on operational timescales.

This regime-based formulation provides the foundation for defining admissible classes of Hamiltonians and for analysing how the quantum-effective sector behaves across families (Σ_N, π_N) , which will be developed in the following sections.

Physical intuition. Approximate projectability states that internal degrees of freedom within each fibre evolve in a way that does not significantly influence observable dynamics over the relevant timescale. The quantum-effective regime therefore corresponds to a form of dynamical decoupling between observable and hidden structure.

4 Admissible Hamiltonian Classes

4.1 Definition of Admissible Hamiltonians

The regime formulation of approximate projectability introduced in the previous section allows one to define a class of Hamiltonians that are compatible with an effective quantum description.

Let $S \subset \Sigma_N$ be a fixed open set with compact closure $\bar{S}$, containing the support of the preparation measure and invariant under the Hamiltonian flow for $t \in [0, T]$, and let $T > 0$ be an operational timescale. A Hamiltonian $H_N : \Sigma_N \rightarrow \mathbb{R}$ is said to be *admissible on (S, T)* if:

1. **(Approximate projectability)** There exists $\varepsilon_N > 0$ such that

$$\sup_{x \in S} \|dH_N|_{V_x}\| \leq \varepsilon_N.$$

2. **(Controlled evolution)** The induced projected dynamics satisfies

$$d_{FS}^{(N)}(\pi_N(x(t)), y(t)) \leq C_T^{(N)} \varepsilon_N,$$

for all $t \in [0, T]$, where $y(t)$ is the effective projected trajectory.

3. **(Regularity)** The Hamiltonian H_N is sufficiently smooth to ensure existence, uniqueness, and stability of solutions over the timescale T .

A Hamiltonian satisfying these conditions is admissible for controlled projected Hamiltonian dynamics. If, in addition, the effective Hamiltonian h_N is operator-induced quadratic in the sense of Definition 2.4, then H_N is Schrödinger-admissible and the projected dynamics lies in the standard finite-dimensional quantum-effective sector.

The admissible class is therefore defined relative to a region S and timescale T . It is not a global property of all Hamiltonians on Σ_N , but a controlled condition specifying when a quantum-effective description is valid.

Admissibility is relative to a triple

$$(S, T, \eta_N),$$

where $S \subseteq \Sigma_N$ is the dynamically relevant region, T is the operational timescale, and η_N is the operational resolution scale on $\mathbb{CP}^{N-1}$. A Hamiltonian is quantum-effective on (S, T, η_N) when the projected deviation induced by its fibre-sensitive component remains below this resolution:

$$C_T^{(N)} \varepsilon_N \ll \eta_N.$$

Thus admissibility is not a purely formal smallness assumption. It is a statement about whether the projected description is accurate at the resolution of the operational experiment.

4.2 Decomposition into Fibre-Invariant and Fibre-Sensitive Components

A useful way to analyse admissibility is through a decomposition of the Hamiltonian into components that are invariant or sensitive to fibre structure.

Given H_N , one may write

$$H_N = h_N \circ \pi_N + \delta H_N,$$

where:

- $h_N : \mathbb{CP}^{N-1} \rightarrow \mathbb{R}$ is an effective Hamiltonian defined on the epistemic space,
- δH_N captures the residual variation along fibres.

The term $h_N \circ \pi_N$ is the component of the ontic Hamiltonian that supports autonomous projected evolution. The residual term δH_N represents fibre-sensitive, coupling-sensitive, or otherwise unresolved structure.

The projected description remains quantum-effective when the vertical derivative of δH_N is small enough that its accumulated projected effect is operationally negligible:

$$C_T^{(N)} \sup_{x \in S} \|d(\delta H_N)_x|_{V_x}\| \ll \eta_N.$$

When this inequality fails, the subsystem is not quantum-effective at the chosen resolution and timescale.

The approximate projectability condition then becomes

$$\sup_{x \in S} \|d(\delta H_N)|_{V_x}\| \leq \varepsilon_N.$$

This decomposition is not unique in general. In the following subsection, we fix a definite representative by conditional fibre averaging, so that the effective Hamiltonian used in the admissibility analysis is no longer arbitrary.

4.2.1 Canonical choice of the effective Hamiltonian

To remove the ambiguity in the decomposition

$$H_N = h_N \circ \pi_N + \delta H_N,$$

we adopt a canonical choice of h_N defined by conditional fibre averaging.

Assume that on the region $S \subset \Sigma_N$, the Liouville measure $\mu_L^{(N)}$ admits a disintegration relative to the projection π_N , with conditional probability measures

$$\{\nu_{[\psi]}^{(N)}\}_{[\psi] \in \pi_N(S)}$$

supported on the fibres

$$F_{[\psi]}^{(N)} \cap S := \pi_N^{-1}([\psi]) \cap S.$$

Define

$$h_N([\psi]) := \int_{F_{[\psi]}^{(N)} \cap S} H_N(x) d\nu_{[\psi]}^{(N)}(x).$$

The fibre-sensitive remainder is then

$$\delta H_N := H_N - h_N \circ \pi_N.$$

By construction,

$$\int_{F_{[\psi]}^{(N)} \cap S} \delta H_N(x) d\nu_{[\psi]}^{(N)}(x) = 0.$$

This construction does not claim uniqueness under arbitrary choices of fibre measure. It fixes a definite representative once a disintegration of $\mu_L^{(N)}$ relative to π_N has been specified.

4.3 Structural Properties of the Admissible Class

The admissible class of Hamiltonians has several important structural features:

Locality in phase space. Admissibility is defined relative to a region S , rather than globally on Σ_N . A Hamiltonian may be admissible in one region and non-admissible in another.

Scale dependence. The parameter ε_N may depend on N , and admissibility is therefore a scale-dependent property. In particular, behaviour that is admissible at small N may fail at larger N , or vice versa.

Timescale dependence. The condition is defined relative to a finite operational timescale T . Over sufficiently long times, even small fibre-sensitive effects may accumulate and invalidate approximate closure.

Non-universality. There is no requirement that all physically relevant Hamiltonians belong to the admissible class. The class identifies a regime of effective quantum behaviour, not a universal law.

4.4 Examples from Finite-Dimensional Models

Finite-dimensional models provide concrete examples of admissible Hamiltonians. In particular:

- Hamiltonians that depend only on the projected state, i.e.

$$H_N = h_N \circ \pi_N,$$

are exactly projectable and therefore trivially admissible.

- Hamiltonians with small perturbations along fibres,

$$H_N = h_N \circ \pi_N + \delta H_N, \quad \|d(\delta H_N)|_{V_x}\| \ll 1,$$

are approximately projectable and admissible over suitable regions and timescales.

- Truncated oscillator and spin models provide explicit cases in which approximate projectability holds and yields effective Schrödinger evolution at the projected level.

These examples illustrate that admissibility is not an abstract condition, but is realised in concrete finite-dimensional systems.

4.5 Non-Admissible Dynamics

Not all Hamiltonians satisfy the admissibility criteria. Failure of approximate projectability may occur when:

- the Hamiltonian varies strongly along fibres,
- fibre-sensitive dynamics couples significantly to horizontal motion,
- or small variations accumulate rapidly over time.

In such cases, projected dynamics becomes ill-defined. Two ontic trajectories that are initially epistemically indistinguishable may evolve into macroscopically distinct projected states, preventing the existence of an effective Hamiltonian description on $\mathbb{CP}^{N-1}$.

Non-admissible dynamics therefore lies outside the quantum-effective regime and cannot be described by standard quantum evolution at the epistemic level.

4.6 Relation to Measurement Contexts and Coupling

Admissibility is also influenced by coupling to external systems, including measurement apparatus and environments. Interactions may either preserve or disrupt approximate projectability.

Preserving interactions. Weak or structured couplings that do not significantly amplify fibre-sensitive variation may maintain admissibility over relevant timescales.

Disruptive interactions. Strong or irregular couplings may increase variation along fibres, leading to breakdown of approximate projectability and loss of effective closure.

This dependence highlights that admissibility is not solely a property of isolated Hamiltonians, but may depend on the broader dynamical context in which the system is embedded.

4.7 Summary

The admissible class of Hamiltonians is defined by controlled approximate projectability on a specified region and timescale. Its key features are:

- a decomposition into fibre-invariant and fibre-sensitive components,
- scale and timescale dependence,

- non-universality, and
- sensitivity to dynamical context.

This class provides the foundation for analysing projected dynamics and for identifying the regime in which finite-dimensional ontic models admit an effective quantum description.

5 Projected Dynamics in the Admissible Regime

5.1 Induced Vector Fields on $\mathbb{CP}^{N-1}$

Let $H_N : \Sigma_N \rightarrow \mathbb{R}$ be an admissible Hamiltonian on a region $S \subset \Sigma_N$. By construction, there exists a function $h_N : \mathbb{CP}^{N-1} \rightarrow \mathbb{R}$ such that

$$H_N \approx h_N \circ \pi_N$$

on S , up to controlled error.

The function h_N induces a Hamiltonian vector field X_{h_N} on $\mathbb{CP}^{N-1}$ with respect to the Fubini–Study symplectic form. The corresponding flow

$$\psi(t)^{(N)} : CP^{N-1} \rightarrow CP^{N-1}$$

provides a candidate effective evolution for projected states.

In the case of exact projectability, the ontic Hamiltonian vector field X_{H_N} is π_N -related to X_{h_N} . In the admissible regime, this relation holds approximately, with deviations controlled by the parameter ε_N .

5.2 Approximate Closure of Epistemic Dynamics

Let $x(t) = \Psi_t^{(N)}(x_0)$ be an ontic trajectory with initial condition $x_0 \in S$, and define the projected trajectory

$$\psi(t) = \pi_N(x(t)).$$

Let $y(t) = \psi(t)^{(N)}(\psi_0)$ be the effective trajectory generated by h_N , with $\psi_0 = \pi_N(x_0)$. Under the admissibility conditions, the two trajectories remain close over finite timescales.

More precisely, for $t \in [0, T]$,

$$d_{FS}^{(N)}(\psi(t), y(t)) \leq C_T^{(N)} \varepsilon_N,$$

where $d_{FS}^{(N)}$ denotes the Fubini–Study distance and $C_T^{(N)}$ depends on the time horizon, projection geometry, and regularity of the flow.

This establishes an *approximate closure* of epistemic dynamics: the projected evolution is well defined up to controlled error and does not depend sensitively on the choice of ontic representative within a fibre.

5.3 Relation to Schrödinger-Form Evolution

On CP^{N-1} , Hamiltonian flow with respect to the Fubini–Study symplectic structure defines a geometric evolution on projective state space (Kibble 1979; Ashtekar and Schilling 1999). Standard finite-dimensional Schrödinger evolution is recovered only when the effective projected Hamiltonian h_N is quadratic, equivalently when it is induced by a Hermitian operator on the underlying Hilbert-space representative.

Accordingly, approximate projectability by itself yields controlled projected Hamiltonian dynamics on CP^{N-1} . Recovery of the standard quantum-effective sector requires the additional condition that h_N lie in the operator-induced quadratic class. In that regime, the projected trajectory is well approximated by Schrödinger-form evolution over finite operational times.

5.4 Domain of Validity

The validity of the approximate closure result depends on several parameters:

- the magnitude of ε_N , controlling variation along fibres,
- the operational timescale T ,

- and the regularity properties of the Hamiltonian flow.

For fixed T , the condition $C_T^{(N)} \varepsilon_N \ll 1$ ensures that deviations from exact projectability remain negligible. However, for longer timescales, small errors may accumulate, leading to breakdown of the effective description.

The admissible regime is therefore defined jointly by (ε_N, T) . It is not an absolute property of the system, but a statement about the validity of the projected description over a specified range of conditions.

5.5 Dependence on Initial Conditions

The approximate closure result is formulated for trajectories with initial conditions in a region $S \subset \Sigma_N$. If the system evolves outside this region, the admissibility conditions may no longer hold.

In particular, regions of phase space with strong fibre-sensitive variation may lead to rapid divergence between $\psi(t)$ and $y(t)$. The effective Hamiltonian description then breaks down, and the system must be described at the ontic level.

This highlights that the quantum-effective regime is not necessarily global, but may depend on the preparation of the system and the portion of phase space it explores.

5.6 Comparison with Exact Projectability

In the case of exact projectability, the relation

$$H_N = h_N \circ \pi_N$$

holds exactly, and the projected dynamics is strictly closed. The diagram relating ontic and epistemic flows commutes, and the induced dynamics on $\mathbb{CP}^{N-1}$ is exact.

The admissible regime generalises this situation by allowing small deviations from exact projectability. Instead of requiring strict commutativity, one requires that deviations remain bounded and operationally negligible.

This generalisation is essential for realistic systems, where exact projectability is unlikely to hold, but approximate closure may still be sufficient to recover quantum behaviour.

5.7 Controlled Effective-Isolation Theorem

The following theorem is the technical core of the paper. It states that if the fibre-sensitive component of the ontic Hamiltonian is bounded, then the projected dynamics remains close to a closed Hamiltonian flow on $\mathbb{CP}^{N-1}$. The result is an effective-isolation theorem because the projected description is valid precisely when the induced deviation is smaller than the operational resolution scale.

Theorem 1. *Let H_N be admissible on (S, T) , with $S \subset \Sigma_N$ a fixed open set with compact closure and with trajectories remaining in S for $t \in [0, T]$. Then there exists an effective Hamiltonian h_N on $\pi_N(S) \subset \mathbb{CP}^{N-1}$ such that, for every $x_0 \in S$,*

$$d_{FS}^{(N)}(\pi_N(x(t)), y(t)) \leq C_T^{(N)} \varepsilon_N, \quad t \in [0, T],$$

where $x(t) = \Psi_t^{(N)}(x_0)$ and $y(t) = \psi(t)^{(N)}(\pi_N(x_0))$ is the effective Hamiltonian trajectory generated by h_N .

If, in addition,

$$C_T^{(N)} \varepsilon_N \ll \eta_N,$$

then the projected dynamics is operationally indistinguishable from closed quantum-effective dynamics at resolution η_N over the interval $[0, T]$.

This theorem is conditional on the admissibility assumptions, the fixed-region hypothesis, and the existence of the effective Hamiltonian h_N . The full hypothesis package and proof are given in Appendix B.

5.8 Illustrative Example: Fibre-Perturbed Hamiltonian

A useful test case is a fibre-perturbed Hamiltonian of the form

$$H_N = h_N \circ \pi_N + \lambda_N K_N,$$

where $h_N \circ \pi_N$ is exactly projectable and K_N is a fibre-sensitive perturbation. Suppose that on the relevant region $S \subseteq \Sigma_N$,

$$\sup_{x \in S} \|dK_N|_{V_x}\| \leq B_N.$$

Then the projectability-error parameter satisfies

$$\epsilon_N \leq |\lambda_N| B_N.$$

The quantum-effective condition becomes

$$C_T^{(N)} |\lambda_N| B_N \ll \eta_N.$$

This example makes explicit how weak residual fibre coupling can support approximate projectability without requiring exact fibre-invariance.

5.9 Summary

Within the admissible class of Hamiltonians, projected dynamics exhibits the following properties:

- the existence of an effective Hamiltonian h_N on $\mathbb{CP}^{N-1}$,
- approximate closure of epistemic trajectories,
- compatibility with Schrödinger-form evolution when h_N lies in the operator-induced quadratic class,
- and controlled deviations determined by ϵ_N and T .

These results establish the dynamical content of the quantum-effective regime and provide the basis for analysing how this regime behaves under scaling to larger systems.

6 Scaling Behaviour and Large- N Structure

6.1 Families (Σ_N, π_N) and Compatibility Conditions

We consider a sequence of finite-dimensional ontic manifolds $\{\Sigma_N\}_{N \in \mathbb{N}}$, each equipped with:

- a Kähler structure $(\omega_\Sigma^{(N)}, J_\Sigma^{(N)}, g_\Sigma^{(N)})$ and associated Liouville measure $\mu_L^{(N)}$,
- a Hamiltonian flow $\Psi_t^{(N)}$,
- and a projection map $\pi_N : \Sigma_N \rightarrow \mathbb{CP}^{N-1}$.

To analyse scaling behaviour, it is necessary to impose compatibility conditions across N . These conditions do not require a literal embedding $\Sigma_N \hookrightarrow \Sigma_{N+1}$, but they do require that the sequence be comparable in a controlled sense.

At minimum, one requires:

1. **Structural compatibility:** The manifolds Σ_N share a common class of geometric structures, allowing comparison of Hamiltonian dynamics across N .
2. **Projection consistency:** The maps π_N behave coherently with respect to changes in dimension, so that epistemic states at level N correspond in a controlled way to those at level $N + 1$.
3. **Observable identification:** There exists a prescription for identifying corresponding observables and measurement contexts across different values of N .

These conditions define what it means to treat (Σ_N, π_N) as a scaling family rather than as unrelated models.

6.2 Behaviour of Approximate Projectability Under Scaling

Let H_N be a sequence of Hamiltonians on Σ_N , each admissible on a region S_N with parameter ϵ_N . The behaviour of approximate projectability under scaling is determined by the sequence $\{\epsilon_N\}$.

Three qualitatively distinct cases arise:

- **Stable regime:** If ϵ_N remains bounded or decreases with N , approximate projectability persists across scales. The quantum-effective regime is stable under scaling.
- **Marginal regime:** If ϵ_N grows slowly with N , admissibility may persist over restricted timescales or regions, but may fail in the limit $N \rightarrow \infty$.

- **Breakdown regime:** If ε_N grows rapidly with N , approximate projectability fails, and no effective projected dynamics exists at large N .

The admissible bridge regime is therefore characterised by sequences $\{H_N\}$ for which ε_N remains sufficiently small relative to the operational timescale.

6.3 Persistence of Admissibility Across Scales

For approximate projectability to support a continuum-like interpretation, admissibility must persist in a controlled way as N increases.

Let T_N denote the operational timescale at level N . A natural condition for persistence is

$$C_{T_N}^{(N)} \varepsilon_N \ll 1$$

uniformly in N , or at least over a range of N relevant to the physical system under consideration.

The constant $C_{T_N}^{(N)}$ is itself part of the scaling data of the family. It depends on the projected residual bound K_N , the Lipschitz constant $L_N(T_N)$ of the effective vector field, and the operational timescale T_N . In favourable cases it may remain uniformly bounded in N . In other cases it may grow polynomially, exponentially, or in a model-dependent manner. The present paper does not derive a universal law for $C_{T_N}^{(N)}$. Instead, bridge admissibility is defined by control of the product $C_{T_N}^{(N)} \varepsilon_N$.

This condition ensures that deviations from exact projectability do not accumulate to the point of invalidating the effective description. It defines a *scaling admissibility criterion* for sequences of Hamiltonians.

6.4 Constraints from Convergence Requirements

In frameworks that analyse large- N limits via convergence of finite-dimensional models, observables and outcome statistics are compared across different values of N . For such comparisons to be meaningful, the underlying dynamics must be sufficiently coherent.

Approximate projectability plays a key role in this context. If projected dynamics is not well defined at each N , then the comparison of epistemic trajectories or distributions across N becomes ill posed.

Accordingly, admissibility provides a necessary condition for convergence analysis: only sequences of Hamiltonians satisfying the scaling admissibility criterion can support a consistent large- N interpretation of dynamics.

6.5 Role of Fibre Structure in Scaling

The behaviour of fibres under scaling is central to understanding approximate projectability at large N . As N increases, both the dimension of Σ_N and the structure of the fibres $\mathcal{F}_\psi^{(N)}$ may change.

Two competing effects may arise:

- **Increased fibre complexity:** Higher-dimensional fibres may allow greater variation of the Hamiltonian along vertical directions, potentially increasing ε_N .
- **Effective averaging:** Larger fibre dimensions may also enhance averaging effects, suppressing the impact of fibre-sensitive variation on projected dynamics.

The net behaviour of ε_N therefore depends on the interplay between these effects. Determining this behaviour requires additional structural assumptions or model-specific analysis, which is not pursued here.

6.6 Limitations of the Scaling Analysis

The scaling framework developed in this section is deliberately conservative. In particular:

- No assumption is made that the sequence Σ_N converges to an infinite-dimensional manifold.
- No claim is made that admissibility persists in the limit $N \rightarrow \infty$.
- No general scaling law for ε_N is derived.

The analysis is therefore restricted to identifying conditions under which approximate projectability can be meaningfully extended across finite-dimensional models.

6.7 Summary

The scaling behaviour of approximate projectability is governed by the sequence $\{\varepsilon_N\}$ and its relation to operational timescales. The key conclusions are:

- admissibility is a scale-dependent property,
- persistence across N requires controlled behaviour of ε_N ,
- and only admissible sequences of Hamiltonians can support a consistent large- N interpretation of projected dynamics.

This establishes the role of approximate projectability as a bridge condition between finite-dimensional models and continuum-like analysis.

7 Relation to the Continuum-Like Bridge

7.1 From Selection Principle to Controlled Condition

In finite-dimensional formulations, approximate projectability functions as a selection principle identifying the quantum-effective regime. The results of the present paper recast this principle as a controlled condition defined in terms of bounded fibre variation and error propagation over operational timescales.

This shift is essential for bridge construction. A selection principle identifies a class of admissible dynamics at fixed dimension, but does not by itself support comparison across scales. A controlled condition, by contrast, provides quantitative criteria that can be evaluated for sequences $\{H_N\}$ and related directly to scaling behaviour.

The admissibility framework developed here therefore replaces an implicit assumption with an explicit regime statement, making it possible to assess when a family of finite-dimensional models can participate in large- N analysis.

7.2 Compatibility with Finite-Dimensional Results

The bridge construction must remain consistent with the finite-dimensional reconstruction established in earlier work. At each fixed N :

- ontic dynamics is given by Hamiltonian flow on Σ_N ,
- observable statistics arise after projection to $\mathbb{CP}^{N-1}$,
- and approximate projectability identifies the regime in which projected dynamics is effectively closed.

The present framework does not alter these results. Instead, it provides a way to organise them across different values of N . In particular, it clarifies how the approximate projectability condition should be interpreted when comparing models at different truncation levels.

7.3 Admissibility as a Bridge Criterion

A central outcome of the analysis is the identification of admissibility as a necessary condition for participation in the continuum-like bridge.

Let $\{H_N\}$ be a sequence of Hamiltonians defined on $\{\Sigma_N\}$. The sequence is said to be *bridge-admissible* if:

1. each H_N is admissible on a region S_N and timescale T_N ,
2. the associated parameters ε_N satisfy a scaling admissibility criterion,
3. and the projected dynamics remains well defined and approximately closed across the sequence.

Only such sequences can support a consistent interpretation of large- N behaviour in terms of effective quantum dynamics.

7.4 Enabling Continuum-Like Interpretation

In approaches that treat continuum behaviour as an asymptotic limit of finite-dimensional models, one compares observables and distributions across increasing values of N . For such comparisons to have dynamical meaning, the underlying evolution must be coherent.

In this sense, $\Sigma 1$ complements the continuum-comparison framework developed in c3. Whereas c3 compares exact finite- N operational statistics across truncation levels, the present paper supplies the corresponding dynamical admissibility condition: only those sequences $\{H_N\}$ whose projected dynamics remains controlled under approximate projectability can participate coherently in later large- N analysis. The two bridge papers therefore address different parts of the same finite-to-large- N transition: c3 controls the comparison of outcome statistics, while $\Sigma 1$ controls the projection behaviour of the underlying ontic dynamics.

Approximate projectability provides this coherence. When admissibility holds, projected trajectories and distributions evolve in a manner that is approximately consistent across scales. This allows one to interpret convergence of observables as reflecting an underlying dynamical structure, rather than as a purely formal limit.

Without admissibility, the projected dynamics at each N may be ill defined or inconsistent, and no meaningful continuum-like interpretation can be assigned.

7.5 Implications for Locality and Structure

Although the present paper does not address locality or field-theoretic structure, the admissibility framework has direct implications for these questions.

Locality and continuum structure are typically formulated in terms of behaviour at large scales or large numbers of degrees of freedom. Any attempt to recover such structure from finite-dimensional models requires a well-defined notion of effective dynamics at each level.

Approximate projectability therefore acts as a prerequisite for such developments. It identifies the class of ontic dynamics for which a coherent large-scale description is possible, without itself specifying the form that locality or spacetime structure should take.

7.6 Relation to Subsequent Developments

The results of this paper are intended to support further work in several directions:

- refinement of convergence results for observables and distributions,
- analysis of composite systems and many-mode structure,
- investigation of locality and effective field behaviour,
- and exploration of possible continuum or spacetime limits.

In each case, admissibility provides the initial filtering criterion: only sequences satisfying the controlled projectability condition are suitable for further analysis.

7.7 Limitations of the Bridge Construction

The bridge framework developed here is deliberately limited. In particular:

- it does not provide a derivation of approximate projectability from deeper principles,
- it does not establish that admissibility persists for all physically relevant systems,
- and it does not construct a continuum theory.

Its role is instead to define the conditions under which such developments may be pursued in a consistent and controlled manner.

7.8 Summary

Approximate projectability, reformulated as a controlled admissibility condition, provides the key link between finite-dimensional models and large- N analysis. It:

- replaces a selection principle with a quantitative regime condition,
- defines a class of bridge-admissible Hamiltonians,
- and establishes the dynamical coherence required for continuum-like interpretation.

This completes the bridge-facing component of the present analysis and prepares the ground for subsequent developments.

8 Scope and Limitations

8.1 What Has Been Established

The present paper provides a structured formulation of approximate projectability as a regime-dependent property of ontic Hamiltonians across a family of finite-dimensional models. The main results may be summarised as follows:

- A definition of approximate projectability as a controlled small-variation condition along fibres of the projection map π_N .
- The introduction of an admissible class of Hamiltonians for which projected dynamics remains well defined up to bounded error over finite operational timescales.
- A formulation of projected dynamics as an approximately closed Hamiltonian flow on CP^{N-1} , compatible with Schrödinger-form evolution when the effective Hamiltonian h_N lies in the operator-induced quadratic class.
- The identification of scaling criteria governing the persistence or breakdown of admissibility across families (Σ_N, π_N) .
- The characterisation of admissibility as a necessary condition for participation in continuum-like or large- N bridge constructions.

These results do not introduce new dynamical postulates. Rather, they reorganise and extend existing finite-dimensional structures into a form suitable for comparison across scales.

8.2 What Has Not Been Established

Several important questions remain open:

- No general derivation of approximate projectability from deeper ontic principles has been provided.
- No universal classification of admissible Hamiltonians has been obtained.
- No scaling law for the behaviour of ε_N has been derived.
- No proof is given that admissibility persists in the limit $N \rightarrow \infty$.

The present paper does not establish any of the following stronger claims:

1. It does not derive the existence of Σ_N or π_N from deeper principles.
2. It does not prove that every physical Hamiltonian is approximately projectable.
3. It does not classify the full set of physically realised projectable or approximately projectable Hamiltonians.
4. It does not classify dynamically realised measurement partitions.
5. It does not derive composite tensor-product kinematics from ontic non-factorisation.
6. It does not derive continuum field theory, locality, or effective field theory.

In particular, the admissibility framework identifies a regime in which effective quantum behaviour is obtained, but does not explain why all physically realised systems should lie within this regime.

8.3 Dependence on Assumptions

The analysis relies on several assumptions that delimit its applicability:

- The existence of a family of finite-dimensional ontic manifolds Σ_N with compatible projection maps π_N .
- Sufficient smoothness of the Hamiltonians H_N to ensure well-defined flows and stability properties.
- The relevance of a bounded region $S \subset \Sigma_N$ and a finite operational timescale T .

These assumptions are natural within the finite-dimensional setting, but may require refinement or modification in more general contexts.

8.4 Status of the quantum-effective projection condition

The present paper improves the status of the quantum-effective projection condition in three layers.

First, it identifies approximate projectability with effective physical isolation. A subsystem is quantum-effective when fibre-sensitive ontic variation remains below operational resolution over the relevant timescale.

Second, it supplies a controlled finite-time theorem: under bounded fibre-sensitive residual dynamics and regularity assumptions,

$$d_{FS}^{(N)}(\pi_N(\Psi_t^{(N)}(x)), y(t)) \leq C_T^{(N)} \epsilon_N.$$

Together with

$$C_T^{(N)} \epsilon_N \ll \eta_N,$$

this gives an operational criterion for closed projected dynamics.

Third, it leaves open the deeper derivation of why physically realised Hamiltonians generically satisfy this condition. That question requires further analysis of the physical mechanisms selecting projectable or approximately projectable Hamiltonians, and is not claimed as closed here.

Thus the condition is not treated merely as a formal selection of favourable Hamiltonians. It is reformulated as a controlled effective-isolation condition. However, a full first-principles derivation remains a later programme target.

8.5 Boundary of Applicability

The admissibility framework is valid only within the regime defined by small fibre variation and controlled error propagation. Outside this regime:

- projected dynamics may fail to be well defined,
- effective Hamiltonian descriptions on $\mathbb{CP}^{N-1}$ may break down,
- and no consistent quantum-effective description is available.

The framework therefore does not apply to systems with strong fibre-sensitive dynamics or to regimes in which small deviations accumulate over long timescales.

8.6 Relation to Continuum and Field-Theoretic Questions

The present analysis does not address continuum, relativistic, or field-theoretic extensions. In particular:

- no infinite-dimensional ontic manifold is constructed,
- no notion of spacetime or locality is derived,
- and no connection to quantum field theory is established.

The admissibility framework should instead be viewed as a prerequisite for such developments. It identifies the class of finite-dimensional models for which a controlled bridge to larger-scale structure may be attempted.

8.7 Open Problems

The results of this paper point to several directions for further work:

- Derivation of approximate projectability from deeper structural or dynamical principles.
- Determination of scaling laws for ε_N in concrete model families.
- Extension of admissibility criteria to composite systems and many-mode structures.
- Investigation of how admissibility interacts with notions of locality and effective field behaviour.

These problems lie beyond the scope of the present paper but are essential for the development of a complete large- N or continuum theory.

8.8 Summary

The framework developed here establishes a controlled regime in which finite-dimensional ontic models admit an effective quantum description. It does so without claiming universality, derivation from first principles, or extension to continuum theories.

Its primary contribution is therefore structural: it defines the admissible conditions under which further developments may proceed in a consistent and quantitatively controlled manner.

9 Conclusion

9.1 Summary of Results

This paper has reformulated approximate projectability as a controlled, regime-dependent property of ontic Hamiltonian dynamics across families of finite-dimensional models. The central result is the identification of an admissible class of Hamiltonians for which projected dynamics on $\mathbb{CP}^{N-1}$ remains well defined and approximately closed over operational timescales.

The analysis establishes that approximate projectability is not a universal structural constraint, but a condition that depends on bounded fibre variation, controlled error propagation, and the relevant dynamical regime. Within this regime, projected dynamics is compatible with Hamiltonian flow on projective state space. Schrödinger-form evolution is recovered when the effective Hamiltonian h_N lies in the operator-induced quadratic class.

By organising these conditions across a scaling family (Σ_N, π_N) , the paper provides a coherent framework for comparing finite-dimensional models and identifying those that support a consistent quantum-effective description at larger N .

The admissibility condition defined here specifies which finite-dimensional ontic dynamics can consistently participate in large- N and continuum-like constructions.

The central result is therefore not a universal derivation of the quantum-effective sector. It is a controlled reformulation of the projectability condition: given (Σ_N, π_N) , quantum-effective dynamics holds when fibre-sensitive ontic variation remains below operational resolution over the relevant timescale. In this sense, approximate projectability is the mathematical form of effective physical isolation.

9.2 Conceptual Implications

The results clarify the status of approximate projectability within the broader theoretical structure. Rather than functioning as an implicit selection principle, it is now expressed as an explicit and testable condition on ontic dynamics.

This shift has two important implications:

- It separates the existence of a quantum-effective regime from the assumption that all dynamics must satisfy it, thereby making the framework more transparent and less restrictive.
- It provides a quantitative basis for analysing when and how effective quantum behaviour emerges from underlying deterministic dynamics.

In this sense, the paper replaces an implicit structural requirement with a controlled approximation framework, making the conditions for effective quantum evolution explicit.

9.3 Role in the Programme

Within the broader research programme, the present work serves as a bridge between finite-dimensional reconstruction and large- N or continuum-like analysis.

Finite-dimensional models establish the operational reproduction of quantum mechanics. The admissibility framework developed here identifies which of those models can be meaningfully extended across scales. It therefore provides the filtering criterion required for subsequent developments, including convergence analysis, many-mode structure, and investigations of locality.

The results do not themselves construct a continuum theory, but they define the conditions under which such a construction may be attempted.

This paper gives a controlled response to the quantum-effective projection problem. That problem requires either a derivation of the quantum-effective sector from deeper ontic principles or a controlled approximation theorem with explicit error bounds. The present paper supplies the second kind of result.

9.4 Limitations and Outlook

The scope of the present work is deliberately limited. No universal derivation of approximate projectability has been given, and no claim is made that all physically realised Hamiltonians belong to the admissible class. The scaling behaviour of the admissibility parameter ε_N remains to be determined in concrete models.

Future work should address these limitations by:

- deriving approximate projectability from deeper structural principles,
- establishing explicit scaling laws in representative systems,
- extending the analysis to composite and many-mode configurations,
- and exploring the emergence of locality and effective field behaviour.

9.5 Final Remarks

The main contribution of this paper is to place approximate projectability on a controlled and explicit footing. By defining an admissible regime for ontic Hamiltonian dynamics, it provides the first bridge-ready formulation of the conditions under which effective quantum evolution arises.

This establishes a foundation for further developments while maintaining a clear distinction between what has been achieved and what remains open.

A Operational Resolution on Projective State Space

A.1 Purpose

The projectability condition produces a geometric deviation bound on $\mathbb{CP}^{N-1}$. To interpret this physically, the deviation must be compared with the resolution of the relevant operational description. This appendix defines the scale η_N used in the main text.

A.2 Resolution from outcome statistics

Let M be a measurement context with outcome probabilities

$$p_M(k|[\psi]).$$

For a statistical distinguishability threshold $\delta_{\text{op}} > 0$, define

$$\eta_N(M, \delta_{\text{op}}) = \inf \left\{ d_{FS}^{(N)}([\psi], [\phi]) : \|p_M(\cdot|[\psi]) - p_M(\cdot|[\phi])\|_{\text{TV}} \geq \delta_{\text{op}} \right\}.$$

Then projected deviations smaller than η_N are operationally unresolved for the context M at threshold δ_{op} .

A.3 Resolution from coarse-grained observables

For a coarse-grained observable family $\mathcal{C}$, one may instead define η_N as the smallest Fubini-Study displacement that changes at least one coarse-grained bin probability by more than δ_{op} . This version is useful when the relevant measurement does not resolve individual projective states but only finite bins or detector regions.

A.4 Interpretation

The condition

$$C_T^{(N)} \epsilon_N \ll \eta_N$$

states that the accumulated projectability error is smaller than the operational resolution of the experiment. This does not imply exact projectability. It implies quantum-effective closure at the specified scale.

B Controlled Projectability Machinery

B.1 Kähler Ontic Setup

At each finite truncation level N , the ontic setup consists of a compact, connected Kähler manifold

$$(\Sigma_N, \omega_\Sigma^{(N)}, J_\Sigma^{(N)}, g_\Sigma^{(N)}),$$

where

$$d\omega_\Sigma^{(N)} = 0, \quad \omega_\Sigma^{(N)} \text{ is non-degenerate,}$$

and

$$g_\Sigma^{(N)}(u, v) = \omega_\Sigma^{(N)}(u, J_\Sigma^{(N)}v).$$

The Liouville measure is the symplectic volume measure

$$\mu_L^{(N)} = \frac{(\omega_\Sigma^{(N)})^{d_N}}{d_N!}, \quad d_N = \frac{1}{2} \dim_{\mathbb{R}} \Sigma_N.$$

This measure is finite because Σ_N is compact.

A Hamiltonian $H_N : \Sigma_N \rightarrow \mathbb{R}$ defines a unique Hamiltonian vector field X_{H_N} by

$$\iota_{X_{H_N}} \omega_\Sigma^{(N)} = dH_N,$$

and hence a deterministic flow

$$\Psi_t^{(N)} : \Sigma_N \rightarrow \Sigma_N$$

which preserves $\mu_L^{(N)}$.

These assumptions define the ontic geometric layer. The projection

$$\pi_N : \Sigma_N \rightarrow CP^{N-1}$$

is extra structure placed on top of the Kähler ontic setup.

B.2 Projection Data

The epistemic projection is a smooth surjective map

$$\pi_N : \Sigma_N \rightarrow CP^{N-1}.$$

For each $x \in \Sigma_N$, the vertical subspace is

$$V_x := \ker(d\pi_N)_x.$$

Fibres are denoted

$$F_{[\psi]}^{(N)} := \pi_N^{-1}([\psi]).$$

When the Controlled Effective-Isolation Theorem is invoked, we additionally assume a smooth horizontal distribution

$$\mathcal{H}_x \subset T_x \Sigma_N$$

such that

$$T_x \Sigma_N = V_x \oplus \mathcal{H}_x.$$

On this horizontal distribution, the projection is assumed to be symplectically compatible:

$$\omega_\Sigma^{(N)}|_{\mathcal{H}_x} = (\pi_N^* \omega_{FS}^{(N)})|_{\mathcal{H}_x}.$$

This compatibility condition is what allows the exactly projectable component $h_N \circ \pi_N$ to push forward to the Hamiltonian vector field generated by h_N on CP^{N-1} .

B.3 Exact Projectability

The strongest dynamical compatibility condition is exact projectability.

Definition B.1 (Exact projectability). A Hamiltonian H_N is exactly projectable on a region $S \subset \Sigma_N$ if there exists a smooth function

$$h_N : \pi_N(S) \rightarrow \mathbb{R}$$

such that

$$H_N|_S = h_N \circ \pi_N.$$

Equivalently, H_N is constant on each connected component of every fibre intersected with S .

If exact projectability holds, the Hamiltonian vector field X_{H_N} is π_N -related to the Hamiltonian vector field X_{h_N} on $\mathbb{CP}^{N-1}$, and the projected dynamics is exactly closed.

B.4 Approximate Projectability

For the bridge problem, exact projectability is too restrictive. The relevant notion is controlled failure of fibre-invariance.

Definition B.2 (Approximate projectability). Let $S \subset \Sigma_N$ be a fixed open set with compact closure $\bar{S}$, chosen before the admissibility analysis and assumed to contain the relevant preparation support. A Hamiltonian H_N is approximately projectable on S with parameter $\varepsilon_N > 0$ if

$$\sup_{x \in S} \|dH_N|_{V_x}\| \leq \varepsilon_N.$$

Thus, approximate projectability requires that the Hamiltonian vary only weakly along the hidden fibre directions. In the limit $\varepsilon_N = 0$, one recovers exact projectability.

A useful equivalent formulation is obtained by writing

$$H_N = h_N \circ \pi_N + \delta H_N,$$

where h_N is an effective Hamiltonian on $\pi_N(S)$ and δH_N measures fibre-sensitive variation. Approximate projectability then becomes

$$\sup_{x \in S} \|d(\delta H_N)|_{V_x}\| \leq \varepsilon_N.$$

B.5 Canonical Choice of Effective Hamiltonian

The effective Hamiltonian h_N is fixed by conditional fibre averaging. Assume that on $S \subset \Sigma_N$, the Liouville measure $\mu_L^{(N)}$ admits a disintegration relative to the projection π_N , with conditional probability measures

$$\{\nu_{[\psi]}^{(N)}\}_{[\psi] \in \pi_N(S)}$$

supported on the fibres

$$F_{[\psi]}^{(N)} \cap S := \pi_N^{-1}([\psi]) \cap S.$$

Define

$$h_N([\psi]) := \int_{F_{[\psi]}^{(N)} \cap S} H_N(x) d\nu_{[\psi]}^{(N)}(x).$$

Then

$$\delta H_N := H_N - h_N \circ \pi_N$$

has vanishing conditional fibre average:

$$\int_{F_{[\psi]}^{(N)} \cap S} \delta H_N(x) d\nu_{[\psi]}^{(N)}(x) = 0.$$

This construction does not claim uniqueness under arbitrary choices of fibre measure. It fixes a definite representative once a disintegration of $\mu_L^{(N)}$ relative to π_N has been specified.

B.6 Controlled Effective-Isolation Theorem

The main bridge statement is conditional rather than universal.

Theorem 2 (Controlled projected dynamics). *Let $S \subset \Sigma_N$ be a fixed open set with compact closure $\overline{S}$, and let H_N be approximately projectable on S with parameter ε_N . Assume:*

1. *the Hamiltonian flow $\Psi_t^{(N)}$ exists on $[0, T]$ and maps the chosen region S into itself over that interval;*
2. *there exists a smooth effective Hamiltonian*

$$h_N : \pi_N(S) \rightarrow \mathbb{R}$$

such that

$$H_N = h_N \circ \pi_N + \delta H_N$$

with

$$\sup_{x \in S} \|d(\delta H_N)|_{V_x}\| \leq \varepsilon_N;$$

3. *the projection π_N is symplectically compatible with the chosen horizontal distribution $\mathcal{H}$, in the sense that*

$$\omega_{\Sigma}^{(N)}|_{\mathcal{H}} = (\pi_N^* \omega_{FS}^{(N)})|_{\mathcal{H}};$$

4. *the projected residual vector field is controlled in the sense that there exists $K_N > 0$ such that*

$$\|(d\pi_N)_x(X_{\delta H_N}(x))\|_{FS} \leq K_N \varepsilon_N$$

for all $x \in S$;

5. *the projected effective flow*

$$\psi(t)^{(N)} : \pi_N(S) \rightarrow \pi_N(S)$$

generated by h_N with respect to $\omega_{FS}^{(N)}$ exists on $[0, T]$.

Then for every initial condition $x_0 \in S$, writing

$$x(t) := \Psi_t^{(N)}(x_0), \quad \psi(t) := \pi_N(x(t)), \quad y(t) := \psi(t)^{(N)}(\psi_0), \quad \psi_0 := \pi_N(x_0),$$

there exists a constant $C_T^{(N)} > 0$, independent of x_0 , such that

$$d_{FS}^{(N)}(\psi(t), y(t)) \leq C_T^{(N)} \varepsilon_N \quad \text{for all } t \in [0, T].$$

Interpretation. The theorem states that if fibre-sensitive Hamiltonian variation is uniformly small on the relevant region, then the true projected ontic trajectory remains close to a genuine Hamiltonian trajectory on projective state space over finite operational times.

B.7 Proof of the Controlled Effective-Isolation Theorem

Proof. Write

$$H_N = h_N \circ \pi_N + \delta H_N$$

on S . Let X_{H_N} , $X_{h_N \circ \pi_N}$, and $X_{\delta H_N}$ denote the corresponding Hamiltonian vector fields. By linearity of the defining relation

$$\iota_{X_F} \omega_{\Sigma}^{(N)} = dF,$$

one has

$$X_{H_N} = X_{h_N \circ \pi_N} + X_{\delta H_N}.$$

For the first term, exact projectability of $h_N \circ \pi_N$, together with horizontal symplectic compatibility of π_N , implies that its pushforward depends only on the projected point and agrees with the Hamiltonian vector field generated by h_N on CP^{N-1} . Hence there exists a Hamiltonian vector field X_{h_N} on $\pi_N(S) \subset CP^{N-1}$, generated by h_N with respect to $\omega_{FS}^{(N)}$, such that

$$(d\pi_N)_x(X_{h_N \circ \pi_N}(x)) = X_{h_N}(\pi_N(x))$$

for all $x \in S$.

It follows that

$$(d\pi_N)_x(X_{H_N}(x)) - X_{h_N}(\pi_N(x)) = (d\pi_N)_x(X_{\delta H_N}(x)).$$

The admissibility hypothesis includes control of the projected residual vector field:

$$\|(d\pi_N)_x(X_{\delta H_N}(x))\|_{FS} \leq K_N \varepsilon_N$$

for all $x \in S$. This is the dynamical error-control condition required in addition to small fibre variation. It ensures that the fibre-sensitive remainder does not generate an uncontrolled horizontal drift after projection.

Therefore

$$\|(d\pi_N)_x(X_{H_N}(x)) - X_{h_N}(\pi_N(x))\|_{FS} \leq K_N \varepsilon_N.$$

Now let

$$x(t) := \Psi_t^{(N)}(x_0), \quad \psi(t) := \pi_N(x(t)),$$

and let

$$y(t) := \psi(t)^{(N)}(\psi_0), \quad \psi_0 := \pi_N(x_0),$$

where $\psi(t)^{(N)}$ is the Hamiltonian flow generated by h_N on $\pi_N(S)$. Since $\pi_N(S)$ has compact closure and X_{h_N} is smooth, there exists a Lipschitz constant $L_N(T)$ for X_{h_N} on the portion of $\pi_N(S)$ explored for $t \in [0, T]$.

Define

$$D(t) := d_{FS}^{(N)}(\psi(t), y(t)).$$

Standard stability estimates for ordinary differential equations on compact Riemannian manifolds give

$$|D'(t)| \leq L_N(T)D(t) + K_N \varepsilon_N$$

for almost every $t \in [0, T]$, with $D(0) = 0$. Applying Grönwall's inequality yields

$$D(t) \leq \frac{K_N}{L_N(T)}(e^{L_N(T)t} - 1)\varepsilon_N$$

when $L_N(T) > 0$, and

$$D(t) \leq K_N t \varepsilon_N$$

when $L_N(T) = 0$.

Hence, defining

$$C_T^{(N)} := \begin{cases} \frac{K_N}{L_N(T)}(e^{L_N(T)T} - 1), & L_N(T) > 0, \\ K_N T, & L_N(T) = 0, \end{cases}$$

we obtain

$$d_{FS}^{(N)}(\psi(t), y(t)) \leq C_T^{(N)} \varepsilon_N \quad \text{for all } t \in [0, T].$$

This proves the theorem. □

B.8 Domain of Validity

The theorem is local in three senses.

Region dependence. Approximate projectability is imposed only on S . If trajectories leave S , the bound need not persist.

Timescale dependence. The constant $C_T^{(N)}$ generally grows with T and may also depend on N . The effective description is therefore controlled only on finite operational timescales.

Scale dependence. In a scaling family (Σ_N, π_N) , both ε_N and $C_T^{(N)}$ may depend on N . The bridge regime is defined by the requirement

$$C_T^{(N)} \varepsilon_N \ll 1.$$

B.9 Why This Appendix Is Conditional

This appendix does not claim that all physically realised Hamiltonians satisfy approximate projectability, nor that a universal derivation from deeper ontic principles has already been achieved. Its role is narrower: to state precisely the machinery needed for the admissible bridge regime and to isolate the controlled condition under which projected Hamiltonian dynamics emerges, with Schrödinger-form dynamics obtained when the effective Hamiltonian belongs to the operator-induced quadratic class.

B.10 Summary

The controlled projectability machinery consists of:

1. a projection map

$$\pi_N : \Sigma_N \rightarrow \mathbb{CP}^{N-1};$$

2. the vertical subspaces

$$V_x = \ker(d\pi_N)_x;$$

3. a small fibre-variation condition

$$\sup_{x \in S} \|dH_N|_{V_x}\| \leq \varepsilon_N;$$

4. an effective Hamiltonian h_N on projective space;

5. and a conditional error bound

$$d_{FS}^{(N)}(\psi(t), y(t)) \leq C_T^{(N)} \varepsilon_N.$$

This is the technical core of the admissible bridge regime used in the main text.

C Large- N Comparison Scaffold and Scaling Data

C.1 Purpose of the Appendix

The main text formulates approximate projectability as a controlled condition at fixed finite N and then asks how that condition behaves across a family of finite-dimensional models. To make that comparison precise, one needs a minimal scaffold specifying what is being compared, how the comparison is performed, and which quantities are required to remain controlled.

The purpose of this appendix is to supply that scaffold. No infinite-dimensional ontic manifold is assumed, and no literal limit object is constructed. The role of the appendix is only to define the finite- N data and the comparison structure needed for a disciplined large- N analysis.

C.2 Scaling Families

A *scaling family* consists of the following data for each $N \in \mathbb{N}$:

1. a compact, connected Kähler ontic manifold

$$(\Sigma_N, \omega_\Sigma^{(N)}, J_\Sigma^{(N)}, g_\Sigma^{(N)});$$

2. the associated Liouville measure $\mu_L^{(N)}$;

3. a smooth surjective projection

$$\pi_N : \Sigma_N \rightarrow \mathbb{CP}^{N-1};$$

4. a Hamiltonian

$$H_N : \Sigma_N \rightarrow \mathbb{R};$$

5. a dynamically relevant region

$$S_N \subset \Sigma_N;$$

6. an operational timescale

$$T_N > 0.$$

The family is denoted

$$\mathcal{F} = \{(\Sigma_N, \omega_\Sigma^{(N)}, J_\Sigma^{(N)}, g_\Sigma^{(N)}, \mu_L^{(N)}, \pi_N, H_N, S_N, T_N)\}_{N \in \mathbb{N}}.$$

C.3 Comparison Principles Across N

The members of a scaling family need not be related by literal embeddings

$$\Sigma_N \hookrightarrow \Sigma_{N+1}.$$

Instead, comparison across N is imposed through three weaker principles.

Definition C.1 (Structural compatibility). A scaling family is structurally compatible if the manifolds Σ_N belong to a common geometric class, the projection maps π_N are defined by a common structural prescription, and the Hamiltonians H_N are specified by a common modelling rule across N .

Definition C.2 (Projection compatibility). A scaling family is projection-compatible if the maps π_N vary coherently with N in the sense that corresponding ontic sectors are sent to corresponding projective sectors under the chosen comparison maps.

Definition C.3 (Observable compatibility). A scaling family is observable-compatible if one has a rule for identifying corresponding preparation families, observables, and measurement contexts across different values of N .

These definitions do not imply that the Σ_N form a nested geometric tower. They require only that the family can be compared in a disciplined way.

C.4 Comparison Maps

To compare different truncation levels, introduce a family of coding or comparison maps

$$\chi_{N \rightarrow M} : \mathcal{O}_N \rightarrow \mathcal{O}_M, \quad N \leq M,$$

where $\mathcal{O}_N$ denotes the class of observables, preparation data, and measurement contexts admitted at level N .

Likewise, when needed, one may introduce comparison maps on the projective side,

$$\kappa_{N \rightarrow M} : \mathbb{CP}^{N-1} \rightarrow \mathbb{CP}^{M-1}, \quad N \leq M,$$

or more modestly a family of correspondences between selected projective regions and observables. The present appendix does not require these maps to be unique or canonical. It requires only that some explicit comparison rule be fixed before asymptotic claims are made.

C.5 Admissible Preparation Families

For each N , let $\mathcal{P}_N$ denote an admissible family of preparations. A preparation may be specified ontically by a Liouville-supported density on S_N , or epistemically by its pushforward under π_N .

A preparation family

$$\{\rho_N\}_{N \in \mathbb{N}}$$

is said to be *comparison-admissible* if:

1. each ρ_N is well defined at level N ;
2. the family is linked by the chosen comparison rule across N ;
3. the family probes a stable physical regime, for example fixed low-lying excitations or fixed coarse observables.

This condition prevents artificial comparisons between unrelated states at different truncation levels.

C.6 Admissible Observable Families

For each N , let $\mathcal{M}_N$ denote an admissible family of measurement contexts or observables. An observable family

$$\{M_N\}_{N \in \mathbb{N}}$$

is *comparison-admissible* if:

1. each M_N is operationally meaningful at level N ;
2. the outcome coding remains stable under the chosen comparison scheme;
3. the induced projective regions or effective Hamiltonians are comparable across N .

The key point is that scaling claims should concern families of observables that represent the same physical question at different truncation levels.

C.7 Scaling Data for Approximate Projectability

Given a scaling family, define for each N the projectability parameter

$$\varepsilon_N := \sup_{x \in S_N} \|dH_N|_{V_x}\|, \quad V_x := \ker(d\pi_N)_x.$$

The corresponding effective projected dynamics is controlled over the operational timescale T_N by a bound of the form

$$d_{FS}^{(N)}(\psi(t), y(t)) \leq C_{T_N}^{(N)} \varepsilon_N, \quad t \in [0, T_N].$$

This motivates the following definition.

Definition C.4 (Scaling admissibility criterion). A sequence of Hamiltonians $\{H_N\}$ is scaling-admissible on the family $\{(S_N, T_N)\}$ if

$$C_{T_N}^{(N)} \varepsilon_N \ll 1$$

throughout the range of N under consideration.

This is the precise large- N bridge condition used in the main text.

C.8 Stable, Marginal, and Breakdown Regimes

The scaling admissibility criterion allows one to distinguish three broad regimes.

Stable regime. If

$$\sup_N C_{T_N}^{(N)} \varepsilon_N < \infty$$

and the product remains uniformly small, then approximate projectability persists across the family.

Marginal regime. If

$$C_{T_N}^{(N)} \varepsilon_N$$

remains small only for restricted windows of N , restricted preparation classes, or restricted timescales, then the bridge is only partial.

Breakdown regime. If

$$C_{T_N}^{(N)} \varepsilon_N$$

ceases to be small, then projected quantum-effective dynamics fails as a controlled description at large N .

These distinctions are qualitative, but they provide a disciplined vocabulary for later model-specific studies.

C.9 What Counts as Large- N Here

The phrase *large- N* is used in a deliberately weak sense. It refers only to comparison across increasingly rich finite-dimensional models. It does not mean that an actual infinite-dimensional ontic manifold has already been constructed, and it does not imply that any $N \rightarrow \infty$ limit exists as a completed geometric object.

Accordingly, all claims in the present paper should be read as claims about families of finite-dimensional models together with explicit comparison rules.

C.10 Non-Claims of the Scaffold

This appendix does not establish any of the following:

1. existence of a universal embedding $\Sigma_N \hookrightarrow \Sigma_{N+1}$;
2. convergence of the ontic manifolds themselves;
3. a general law for the asymptotic behaviour of ε_N ;
4. emergence of locality, spacetime, or field structure.

Those questions require additional hypotheses and belong to later work.

C.11 Why This Scaffold Is Sufficient for the Present Paper

The present paper needs only a disciplined way to compare:

- the ontic manifolds Σ_N ,
- the projections π_N ,
- the Hamiltonians H_N ,
- the dynamically relevant regions S_N ,
- and the operational timescales T_N .

Once these data are fixed, approximate projectability can be evaluated at each N , and scaling-admissibility can be stated without overreaching beyond the finite-dimensional regime.

C.12 Summary

The large- N comparison scaffold consists of:

1. a scaling family

$$\{(\Sigma_N, \omega_{\Sigma}^{(N)}, \mu_L^{(N)}, \pi_N, H_N, S_N, T_N)\}_{N \in \mathbb{N}};$$
2. comparison rules for preparations, observables, and projective data;
3. the projectability parameters ε_N ;
4. the control factors $C_{T_N}^{(N)}$;
5. and the scaling admissibility condition

$$C_{T_N}^{(N)} \varepsilon_N \ll 1.$$

This is enough to formulate a disciplined bridge from fixed finite- N projectability to controlled large- N analysis, while leaving stronger continuum and spacetime claims explicitly open.

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